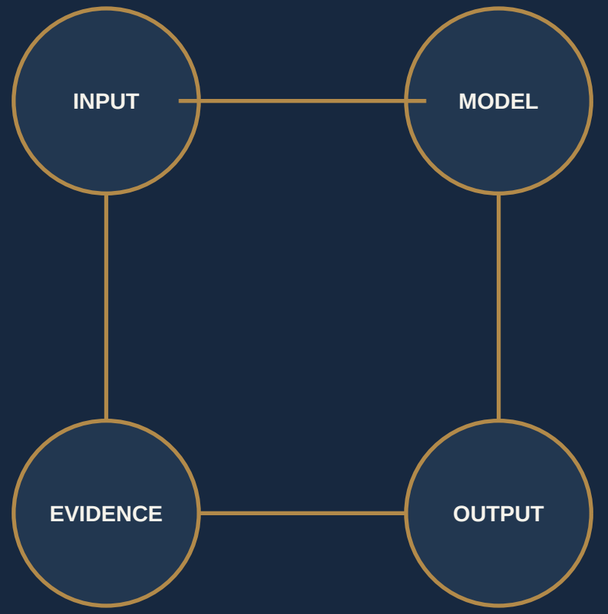
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Multi-Robot Systems

AI.309 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define coordination.
- Apply key methods including communication.
- Build or analyze a practical system involving task allocation.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Coordination

02

METHODS

Communication

03

SYSTEMS

Task allocation

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Coordination

Coordination

02 · Communication

Communication

03 · Task allocation

Task allocation

04 · Distributed planning

Distributed planning

05 · Cooperation and conflict resol

Cooperation and conflict resolution

06 · Mandatory practical: Develop a

Mandatory practical: Develop a multi-robot simulation performing a coordinated mission

Core ideas, made explicit

AI.309 • Distinguish definitions, assumptions and evidence

- Coordination
- Communication
- Task allocation
- Distributed planning
- Cooperation and conflict resolution

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

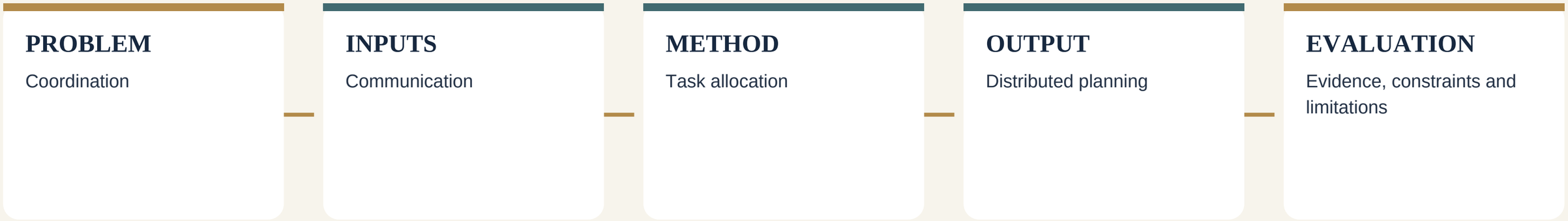
AI.309 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Develop a multi-robot simulation performing a coordinated mission

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.



Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving coordination.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using coordination.
- Implement or simulate a baseline using communication.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.